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Syntax and Pragmatics of Adversativity: Towards a grammar of co-speech gestures*

Abstract

In this article, we examine adversative special questions in Italian expressing disapproval. Our goal is to explore, through an experimental setup, the syntactic triggers involved in the characteristic gestuality associated with these sentences. We argue, building on our previous work, that gestures, particularly non-lexical co-speech gestures, are not just an additional communication channel. Instead, they are a crucial part of the language system, generated and coordinated by the same syntactic principles that govern word order and sentence structure. We will show that expressive language not only triggers a significantly larger number of gestures compared to non-expressive language but also, more interestingly, that each trigger appearing in the syntactic representation is associated with a specific gesture realization. We will demonstrate that the projections needed in the syntax to represent adversativity, interrogativity, and emotionality each trigger a characteristic and predictable gestural pattern. Ultimately, our observations highlight a fundamental multimodal architecture of language, where the alignment of syntax and sensorimotor realizations, including gestures, is a core property of the language model.

Keywords:

syntax-gesture interface, expressive language, co-speech gesture, disapproval, adversativity.

1 Introduction

In this study, we examine adversative special questions in Italian when they convey an emotional value, and precisely disapproval. This focus is motivated by the observation that expressive language reveals a lot about how syntactic structures connect with prosody and gestures, especially since their

* The authors developed all parts of this research together; however, regarding legal responsibility, Alessandra Giorgi takes official responsibility for sections 1, 2, 3, 4 and 8. Erika Petrocchi takes official responsibility for sections 5, 6 and 7. We thank the students who completed their internships at Ca' Foscari under Alessandra Giorgi's supervision and participated in this research: Ester Ferrari, and Manuel Pittelli.

realization differs significantly from non-emotional sentences. Besides providing new insights into the syntax of adversativity, the goal of this article is to explore, through an experimental setup, the syntactic triggers involved in the characteristic gestuality associated with these sentences. We argue, building on our previous work – see Giorgi & Petrocchi (2024, 2025a, 2025b) – that gestures, particularly non-lexical co-speech gestures, are not just an additional communication channel. Instead, they are a crucial part of the language system, generated and coordinated by the same syntactic principles governing word order and sentence structure, making them an essential component of grammar.

We will demonstrate that expressive language not only triggers a significantly larger number of gestures compared to non-expressive language, but also, more interestingly, that each trigger appearing in the syntactic representation is associated with a specific gesture realization. In other words, every functional category present in the syntax for correctly representing adversative special questions can be realized by means of a specific gesture, consistently across speakers.¹

Therefore, even if they lack a *lexical meaning* as traditionally understood, co-speech gestures can systematically convey expressive content, showing that they are part of the linguistic inventory alongside lexical and phonological units.²

This article is organized as follows: in the next section, we will outline the basic theoretical concepts on which our work is based; in section 3, we will discuss the main theoretical issues involved in the analysis of the adversative particle *ma*; in section 4, we will outline our research questions; in section 5, we will present our experimental plan and the materials employed; in section 6, we will provide the statistical analysis of the experiment, and in section 7, we will discuss the theoretical implications of our findings. Finally, in section 8, we will draw some general conclusions.

2 The Theoretical Framework

The theoretical background of this work is constituted by the generative *Minimalist* framework, originally proposed by Chomsky (1995) and further developed by Chomsky (2000, 2001, 2008, 2013) and many other scholars. Even if the generative approach to language does not have the goal of studying pragmatics and gestures in particular, we believe that the theoretical and conceptual

¹ This is reminiscent of the findings by Hinterhölzl and Frascarelli (2007), who pointed out that topic phrases appearing in the Italian left periphery fulfill different discursive functions and are consistently associated with different prosodic realizations. In other words, in the cases they examined prosody is playing an important role to signal the pragmatic function of left-peripheral phrases in Italian, in a way similar to gestures in our study.

² As we will better discuss in the text, gestures are not *obligatory* component of the linguistic realization; (non-sign) language in fact privileges the oral channel, which provides indeed the main clue to identify clause types and expressive properties. Yet, they appear to be systematic and consistent.

apparatus it provides can offer important insights into these fields as well.³ According to this theoretical proposal, the human language faculty is organized with *syntax* as the main component, connecting sound and meaning through the *conceptual component* on one side, which provides the interpretation, and the *sensorimotor apparatus* on the other, which determines sound (and gesture) realization. This model does not allow a direct link between sound and meaning; instead, this relationship must be mediated by the syntactic representation. As an exemplification, consider the following picture:⁴

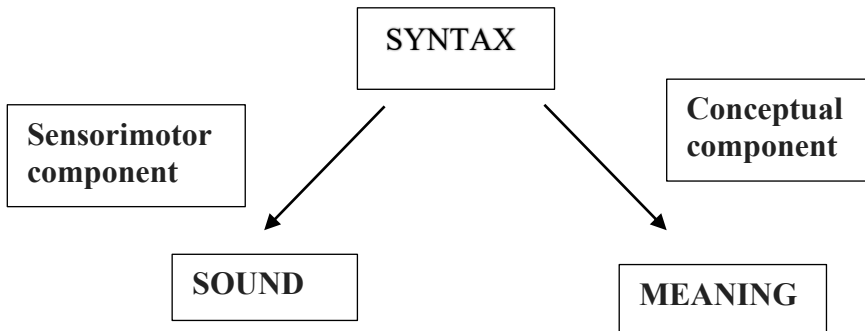


Figure 1: The Minimalist model

As discussed in Giorgi (2023), this approach, compared to a model that allows both a syntax-mediated path and a direct sound-to-meaning path, is theoretically superior because it is more constrained. A system with both paths in fact would be inherently more complex and powerful. Hence, the framework proposed here can be taken to be the null hypothesis.

The implication of this proposal for our discussion is that all features produced by the sensorimotor apparatus must be represented in the syntax. Therefore, both intonation, as far as it constitutes a property distinctive of meaning, and the gestural aspects systematically linked to it, should also be triggered by syntax.⁵

Note that this view is not new in generative grammar. Consider the following example, also discussed in Giorgi (2023):

(1) pro ha telefonato

³ For gesture studies in a generative framework, see among the others the works by Schlenker (2018, to appear), Sailor & Colasanti (2020), Colasanti (2023), Ippolito et al. (2022).

⁴ *Syntax* is to be understood here as *core syntax*, or *narrow syntax*, which is the minimal representation common to every language. In this sense, it roughly corresponds to the notion of *Universal Grammar* as developed in the *Government and Binding* framework (Chomsky, 1981).

⁵ For recent analyses of the syntax/pragmatics interface along these lines, see the special number of the *Journal of Pragmatics* edited by Trotzke (2024) and articles published there. See also Domaneschi & Bambini (2021).

Lit: has called

‘He called’

‘Did he call?’

This sentence can be uttered with two different intonations: a *flat* one, i.e., a falling tone, which makes it an declarative sentence, and a rising tone, which indicates a polar question. In other words, in this case, intonation creates a minimal pair, crucially distinguishing the two interpretations. We might say, therefore, that in this case the intonation *disambiguates* the two readings, apparently suggesting the existence of a direct link between sound and meaning. However, since the very beginning of generative grammar development—see, for instance, Ross (1967) and Chomsky (1975) – it was hypothesized that there is a cue within the syntax that creates the different pairing of sound and meaning. The structure assigned to interrogative sentences includes, in fact, besides a null pronominal subject, which is available in Italian and in so-called *pro-drop* languages, an empty operator, as shown in the following representation:

(2) [_C, OP [_T, pro ha telefonato]]

has called

There are many arguments in favor of this hypothesis, which we are not able to review here. Let us only briefly mention that in this way the representation of polar questions is assimilated to the one provided for the open ones, the only difference being that in open questions the operator is overtly expressed as a *wh*-phrase. Consider for instance the following example:

(3) [_C, Cosa_i [_T, pro ha mangiato *e_i*]]

Lit: what has eaten

‘What did he eat?’

In summary, from a minimalist perspective, it can be said that the syntactic representation in (2) is read off at the interface with the sensorimotor component and that the empty operator triggers the characteristic intonation of polar questions. At the same time, the conceptual component interprets this structure as a question rather than a statement.

In line with this view, we argue for a model of language as a fully integrated multimodal system. We propose that syntax, prosody, and gesture are not independent but inherently connected. Our main idea is that specific syntactic structures automatically trigger corresponding sensorimotor realization.

We will demonstrate in this article that this realization includes not only the expected phonology and intonation but also manual and non-manual gestures.⁶

3 The syntactic properties of the adversative particle *ma*

In this section, we will review the evidence concerning the syntactic properties of the adversative particle *ma*. The goal is to assign the appropriate label to this particle and highlight its properties in the syntactic structure.

3.1 Some general considerations on adversativity

Adversativity across languages is not just a single phenomenon but a constellation of related phenomena. The term itself signifies a contrast, such as A versus B, and different languages have various lexical items and syntactic means to express these contrasts, which are not easily translated from one language to another because of different interpretive nuances.

For instance, in English, there are many ways to convey an adversative meaning, as in the following examples:

- (4) Mary is wealthy, *but* she is not happy.
- (5) *Although* Mary is wealthy, she is not happy.
- (6) Mary is wealthy; *however*, she is not happy.
- (7) Mary is wealthy, *yet* she is not happy.

This list is certainly not exhaustive, but it clearly shows that there are different lexical options to contrast A, Mary being wealthy, and B, Mary being unhappy. Additionally, sentences (4)-(7) might convey slightly different meanings, so the contexts in which the adversative lexical items appear might not be exactly the same. In the technical literature on this topic, the contrast between A and B presented above is called a *counter-expectational* one – see, among others, Vicente (2010) – because it indicates a violation of the speaker's expectations. In this case, the speaker's expectations are based on common knowledge about the implications of being wealthy in our culture, where wealth is expected to automatically lead to happiness.

However, if we consider the distribution of the lexical item *but*, in English, we see that it can appear in sentences where this counter-expectational value is missing, as in the following case:

⁶ The same framework and general hypothesis was also empirically and theoretically discussed in previous works: Giorgi & Dal Farra (2019), Petrocchi (2022), Giorgi & Petrocchi (2024, 2005a, 2025b).

(8) Mary's dress is not blue *but* a dark shade of green.

In this case, there is no expectation that a specific dress must be green or not, which is why these sentences are called *corrective* adversatives instead of counter-expectational. However, this does not mean that there cannot be an expectation about Mary's dress color, as in the following example:

(9) Mary told me she would wear a blue dress. When I saw that her dress was not blue *but* a dark shade of green, I felt disappointed.

Determining whether the sentence in (9) is counter-expectational or corrective is not straightforward. Interestingly, both in (8) and (9), it is impossible to use *although*, *however*, or *yet*. However, it remains unclear whether these incompatibilities can serve as tests to differentiate counter-expectational cases from corrective ones. Given the interpretation of (9) – that I expected Mary's dress to be blue and felt disappointed to find it green – it seems that the definitional issue needs more clarification, and any proposed answer is likely to be non-trivial. We will not examine this question in detail here. Instead, we generally classify any contrast between A and B expressed with *ma* as *adversative*. This is a simplification of a complex issue, but it effectively functions as an operational definition for our purposes. Furthermore, we will limit our analysis to structures where both parts of the adversative construction are full clauses, leaving aside cases involving elliptical structures.

As noted above, languages differ in the way they express adversativity. For instance, Vicente (2010) pointed out that in example (4) – i.e., in counter-expectational cases – Spanish would use the adversative lexical item *pero* (but), whereas in the corrective one in (8), the lexical item *sino* (but) would appear.⁷

In Italian, the lexical item *ma* (but) appears in both counter-expectational and corrective cases:

- (10) Maria è ricca, *ma* non è felice
Maria is wealthy but she is not happy
- (11) Il vestito di Maria non è blu, *ma* verde scuro
Maria's dress is not blue, but dark green

⁷ On Spanish see also García Murga (2017). Note that languages vary a lot in the way they lexicalize adversativity. For instance, Malchukov (2004) discusses four different ways to express adversativity in Russian; see also Kapatsinski (2009).

Even in Italian, there are various ways to express adversativity. For example, as shown in (12) it is possible to substitute *ma* appearing in example (10) with *però*, maintaining the same counter-expectational value:⁸

- (12) Maria è ricca, *però* non è felice
 Maria is wealthy, but she is not happy

The adversative particle *ma* in (11) can also be substituted by *bensi*, as in the following case:

- (13) Il vestito di Maria non è blu, *bensi* verde scuro
 Maria's dress is not blue, but dark green

Bensi, is not available in (12), and *però* is not possible in (13).⁹ We will not discuss here the possible alternative ways to express adversativity across languages, since our focus is on Italian, specifically on the usage of *ma* when introducing full clauses, as we will illustrate in the next section.

Concluding this brief section, we can say that the distinction between counterfactual and corrective adversatives might not be entirely accurate or complete, since each language selects various lexical items to express different nuances and the two interpretations might not be easily distinguished from each other. In this work, instead of focusing on classification issues, we will emphasize the syntactic and semantic properties of disapproval questions introduced by *ma*, as illustrated in the following sections.

3.2 The syntax of *ma*

In this section, we provide a brief illustration of the syntax of the adversative particle '*ma*' in Italian. We consider two aspects: the first issue concerns the labeling, i.e., the category to which this item can be assigned, and the second concerns its distribution.¹⁰

3.2.1 *Ma* is not an adverb: a comparison between *ma* and *però*

As noted earlier, in Italian, adversativity can be expressed in different ways. To determine the

⁸ For an analysis of *però* and other connectives in Italian see also Mauri & Giacalone Ramat (2012).

⁹ See also the discussion in section 3.2.1.

¹⁰ The traditional term used for this item, *coordinating conjunction*, is inadequate, especially because there are many distributional differences between *ma* and the coordinating particles *e* (and), or *o* (or), beyond their lexical contribution. We will not discuss in detail the issues related to comparing these particles but will only highlight some properties that might help us to better understand adversativity.

category of *ma*, we will set aside the properties of *bensi* due to its limited use and briefly examine *ma* and *però*. Unlike Spanish, as described by Vicente (2010), these two items are not clearly linked to different meanings, as we will show soon. The distribution of *però* is typical of high adverbs – see Cinque (1999) – which are base-generated in the left periphery of the clause and can then appear in different positions due to the raising and reordering of other constituents around them. Consider the following examples:

- (14) Maria è astemia, *però* ieri ha bevuto una birra
- (15) Maria è astemia, ieri *però* ha bevuto una birra
- (16) Maria è astemia, ieri ha *però* bevuto una birra
- (17) Maria è astemia, ieri ha bevuto una birra, *però*
Maria is a teetotaler, but yesterday she drank a beer.

This distribution is the same for left-peripheral adverbs, as illustrated in the following examples:

- (18) *Probabilmente* ieri ha bevuto una birra
- (19) Ieri *probabilmente* ha bevuto una birra
- (20) Ieri ha *probabilmente* bevuto una birra
- (21) Ieri ha bevuto una birra, *probabilmente*
Probably yesterday she drank a beer.

Note that the final position of the adverb is only available with a parenthetical reading, as discussed in Giorgi (2018). In contrast, all clause-internal positions are ungrammatical with *ma*, as can be seen in the following cases:

- (22) *Ma* ieri ha bevuto una birra
- (23) *Ieri *ma* ha bevuto una birra
- (24) *Ieri ha *ma* bevuto una birra
- (25) *Ieri ha bevuto una birra, *ma*
But yesterday she drank a beer.

Based on this evidence, we can conclude that *però* is an adverb, while *ma* is not. Also, note that precisely for this reason, the two can appear in the same sentence, even if it might be considered non-polished Italian.

- (26) Maria è astemia, *ma* ieri *però* ha bevuto una birra
 Maria is a teetotaler, *ma* yesterday *però* she drank a beer.
 ‘Maria is a teetotaler, but yesterday she drank a beer.’

Therefore, even if *ma* and *però* express very similar meanings, they do not belong to the same category, given that they do not share the same distribution. Concluding, the adversative particle *ma* cannot be labeled as an adverb; in the following section we will present the proposal we are going to assume here.

3.2.2 *Ma* as a Discourse Head

As discussed by Giorgi (2016, 2018), we can check whether *ma* belongs to the clause's left periphery. Let's analyze the distribution of *ma* with respect to Focus phrases (in capital letters):

- (27) Ma A GIANNI hai comprato un libro (non a Mario)!
 But to Gianni.Foc you bought a book (not to Mario)!
- (28) *A GIANNI *ma* hai comprato un libro (non a Mario)!
 *To Gianni.Focus but you bought a book (not Mario)!
 ‘But you bought a book to Gianni (not to Mario)!’

These examples show that a Focus phrase – in this case, a corrective focus – cannot precede the adversative particle.¹¹ Let's further examine the distribution of Clitic Left Dislocation (CLLD):¹²

- (29) Ma a Gianni gli hai comprato un libro
 But to Gianni.top you to him-bought a book
- (30) *A Gianni *ma* gli hai comprato un libro
 *To Gianni.top but you to him-bought a book
 ‘But you bought a book to Gianni’

The ungrammaticality of the example in (30) shows that CLLD cannot precede *ma*. Following Rizzi

¹¹ Recall that in Italian, as in many languages, a corrective or contrastive focus is associated to a peculiar emphatic focus intonation.

¹² CLLD in Italian is associated to a characteristic intonation; often it is realized with a *comma intonation* typical of parenthetical (Selkirk, 2005). For a discussion, see Giorgi (2015).

(1997), we can argue that CLLD and Focus phrases are located in the syntactic layer known as the *left periphery* of the clause. The data above show that *ma* must precede the items realized in this syntactic layer.

According to Rizzi (1997), on the left of Focus and Topic phrases, there is the high complementizer *che*, which he labels *Force*. We might wonder, therefore, if *ma* can be assimilated to such a complementizer. Consider the following examples:

- (31) *Gianni ha detto che *ma* Maria ha telefonato
Gianni said that *ma* Maria has called
- (32) *Gianni ha detto *ma* che Maria ha telefonato
Gianni said *ma* that Maria has called
- (33) *Gianni ha detto *ma* Maria ha telefonato
Gianni said *ma* Maria called
'Gianni said that Maria called'

The ungrammaticality of the examples (31) and (32) shows that *ma* cannot appear in subordinate clauses with *che*, and (33) shows that it cannot introduce a complement clause; hence, *ma* is not a complementizer, even if it precedes topic and focus as illustrated in (27) – (30).

Following Giorgi (2016, 2018), we hypothesize that *ma* is a discourse head, leading to the following structure:

- (34) [DISC-P [CP]₁ *ma*_{DISC°} [CP]₂]

Several other observations support this characterization of *ma*. Let's go back to example (10) above, reproduced here for clarity:

- (35) Maria è ricca, ma non è felice
Maria is wealthy, but she is not happy

Interestingly, example (35) could also be uttered by two different speakers, as in the following dialogue:

- (36) Speaker A: Maria è ricca

- Maria is wealthy
- (37) Speaker B: *Ma* non è felice
 But she is not happy

In this mini-dialogue, the two parts connected by "*ma*" realize two different speech acts, produced by different speakers. Furthermore, in the same vein, the two parts of a structure linked by "*ma*" can belong to different sentence types.

- (38) Gianni è in salotto, *ma* Maria dov'è?
 Gianni is in the living room, but where is Mary?

In (38) an assertion is combined with a question. This is not limited to interrogative clauses. In (39), the speaker combines an assertion with an order expressed as an imperative:

- (39) Sei entrato qui di nascosto, *ma* adesso vattene!
 You came in here secretly but now go away!

It is a well-established generalization in generative grammar – see, for example, Sadock and Zwicky (1985) – that each sentence can be assigned to only one type, and no sentence can belong to more than one type. This observation, therefore, provides strong evidence in favor of treating the two parts connected by *ma* as a discourse and not as a single sentence.¹³

Note finally that when the first clause contains a sentential negation, the clause introduced by *ma* is always outside its scope:¹⁴

- (40) Gianni non è alto, *ma* gioca a basket
 Gianni is not tall, but he plays basketball
- (41) Gianni non si è comprato una nuova auto, *ma* ha deciso di usare solo i mezzi
 pubblici
 Gianni did not buy a new car, but decided to use only public transportation

¹³ For further discussion, see among the others, König & Siemund (2007).

¹⁴ On the possibility of subclausal coordination with *ma*, see Vicente (2010), and Franco (2016); see also Jasinkaja (2012) for a comparison between adversativity and coordination. A discussion of these cases lies outside the scope of our work, since here we are only considering full clauses. However, we believe that even in apparently subclausal cases, a full clausal analysis with ellipsis could be maintained. This issue will be explored in further work

In both cases, the sentence introduced by *ma* is not affected by negation. Note that in sentence (40) the adversative clause could be classified as *counter-expectational*, whereas in (41) it might be considered a *corrective* one. However, as noted above, this distinction may be purely descriptive, as it is unclear whether it has theoretical relevance.

Finally, consider that *ma* can introduce parentheticals:

- (42) Gianni, *ma* forse non lo sapevi, è laureato in fisica.
Gianni, *but* maybe you didn't know it, graduated in physics.

Parentheticals, as discussed in the literature – see, among others, Dehé and Kavalova (2016) – are largely independent of the host sentence; therefore, in this case, the structure introduced by *ma* is an autonomous linguistic unit; therefore, characterizing *ma* as a discourse particle aligns well with this observation.

Concluding this section, we can say that *ma* is realized on the left of the left periphery, obligatorily preceding both Focus and CLLD, and introduces sentential structures that are independent from the one on its left, as shown by the tests provided above. These tests include the availability of different speakers, the possibility of combining different sentential types, the impossibility of being interpreted within the scope of a negation in the previous sentence, and the possibility of introducing parentheticals. Hence, *ma* can be labeled as a discourse head *DISC*, projecting its own constituent. The sentences on the right and on the left, therefore, form a discourse, or, in the terminology proposed by Giorgi (2023), a *micro-discourse*, i.e., a discourse disguised as a single sentence.

3.2.3 Adversative *ma* in special questions

In this section, we will analyze the properties of *ma* when introducing the so-called *special questions*, a subtype of rhetorical questions that express the speaker's feelings, which is the context we are going to test experimentally.¹⁵

As discussed in our previous work – see Giorgi (2016, 2018), Giorgi & Dal Farra (2019), Petrocchi (2022), Giorgi & Petrocchi (2024) – *ma* can introduce, in the absence of a linguistic context, surprise and disapproval special questions. Consider the following example for surprise, from Giorgi &

¹⁵ On special questions see Munaro and Obenauer (1999), Obenauer (2004, 2006), Bayer and Obenauer (2011), Munaro and Poletto (2003), Obenauer and Poletto (2000), Hinterhölzl and Munaro (2015). Obenauer (2004, 2006) distinguishes special questions from the rhetorical ones, but as argued in the text, they have several properties in common. The issue however, is largely just a terminological one, therefore we will not pursue it here any further,

Petrocchi (2024, ex.1):¹⁶

- (43) Scenario: I know that you are on a diet and decided to eat only fruit. One day I see you eating a big hamburger. I am surprised and utter:
- (44) Ma non mangiavi solo frutta?
But not eat.IMPF.2PS only fruit?
'But weren't you eating only fruit?'

Consider also the following scenario for disapproval questions, from Giorgi & Petrocchi (2024, ex. 2):

- (45) Scenario: I see Gianni wearing his best trousers, kneeling in the dirt in the garden. I think he will ruin his trousers. I am annoyed and utter:
- (46) Ma cosa fai?!
But what do.PRES.2PS
'But what are you doing?!'

In these sentences, the speaker's expectations are not met, leading to surprise and/or disapproval. Similar to rhetorical questions, when uttering them, the speaker is not seeking an answer. Instead, these questions are meant to express her feelings and ask for an explanation of the surprising or disappointing situation. For example, in the scenario described above, the conversation might continue with the interlocutor saying something along these lines:

- (47) Mi dispiace, mi sono scordato di cambiarmi.
I am sorry, I forgot to change.

These sentences can be uttered out of the blue, i.e., without a previous interaction between the speaker and her interlocutor. To account for these constructions, we take *ma* to be a Discourse Head, connecting two independent sentential units as proposed in (34), repeated here for clarity:

- (48) [DISC-P [CP]₁ *ma*_{DISC°} [CP]₂]

¹⁶ The abbreviation IMPF and PRES in the glosses stand for *imperfect* and *present* respectively. This is an interesting feature differentiating surprise from disapproval questions, as discussed in Giorgi (2016, 2018). We will not pursue this issue here any further.

We assume here that the CP on the left, that is, CP₁, remains unexpressed and is constituted by the speaker's expectations set by the previous context. To be more precise, in those cases, the structure is the following (where capital letters represent the unspoken portion):

$$(49) \quad [_{\text{DISC-P}} [\text{SPEAKER EXPECTS } X]_1 \text{ } ma_{\text{DISC}^\circ} [\text{NOT } X]_2]$$

In this example, similarly to the counter-expectational case discussed above – *Gianni è alto, ma non gioca a basket* (Gianni is tall, but he does not play basketball) – where C1 is *Gianni è alto* (Gianni is tall), C1 is THE SPEAKER EXPECTS GIANNI TO EAT ONLY FRUIT, and C2 contradicts that expectation. In other words, the adversative head, by introducing expectations, creates a contrast between C1 and C2. In the following sections, we will illustrate the experiment we set up to better account for the syntax of adversativity.

In the expressive examples provided above – (44) and (46) – other heads must be projected in the syntax, as argued in our previous works. In those cases, in fact, C₂ is both an interrogative and an evaluative sentence. Following Cinque (1999), we hypothesize, therefore, the existence of a high Evaluative (EVAL) projection having scope on the sentence, and an interrogative (INT) projection.¹⁷ The reciprocal order of INT and EVAL is INT > EVAL, as discussed in Giorgi & Dal Farra (2019) and Giorgi & Poletto (in prep). Giorgi and Dal Farra (2019) observed that some of the participants, when asked to repeat a disapproval question such as (34), introduced an explicit negative evaluative word, following the *wh*-:

- (50) Ma cosa *cavolo* fai?
 Lit: But what *cabbage* are you doing?
 ‘What the fuck are you doing?’

In (50), the term *cavolo* functions as an evaluative element that frames the speaker's utterance as expressing a negative assessment – in the scenario provided above, it signals disapproval toward the interlocutor's behavior. The obligatory syntactic order is *ma* (adversative) > INT (interrogative) > EVAL (evaluative term). This sequence, therefore, gives rise to an *adversative interrogative disapproval expression*, where *ma* introduces the contrast, the interrogative item signals it as a question, and the evaluative projection, which may or may not be lexically realized, expresses disapproval.

¹⁷ On the position of the interrogative projection in the left periphery, see Rizzi (2001).

Concluding, the structure we propose for disapproval special questions is as follows, where TP stands for Tense Phrase:

$$(51) \quad [_{\text{DISC-P}} [_{\text{C1}} \text{SP EXPECTS } X] \text{ } ma_{\text{DISC}^\circ} [_{\text{C2}} [_{\text{INT-P}} \text{INT}^\circ [_{\text{EVAL-P}} \text{EVAL}^\circ [_{\text{TP}} \text{NEG } X]]]]]$$

The structure in (51) represents a discourse headed by *ma*, where C1 represents the (unmet) speaker's expectations, and C2 is an interrogative evaluative sentence.

4. Research questions and hypotheses

On the basis of the structure provided above, we can now discuss our research questions and hypotheses. In previous research, we found that the repetition test shows that most participants accompany the production of surprise and disapproval adversative questions with typical intonation and gestures. We also discovered that intonation and gestures align with syntax. Specifically, gestures are aligned with the left periphery of the clause, with their onset being either simultaneous with or just before the phonological realization. We also found that the gesture repertoire produced in these sentences is quite uniform across languages, pointing therefore to a deeper motivation for their occurrence.¹⁸

Note that non-lexical co-speech gestures do not immediately serve a grammatical role. They are consistently present and triggered uniformly across different languages and speakers, but apparently, they lack a specific function: Giorgi and Dal Farra (2019) experimentally observed that co-speech gestures are produced even without an interlocutor – for instance, when speaking on the phone – which shows that the appropriate pairing of sound and meaning occurs even without visual access to gestural cues. The very fact that co-speech gestures are present, even without a clear function, appears to be a fundamental aspect of language, indicating that language is inherently multimodal, as argued by Giorgi and Petrocchi (2025a). Therefore, exploring how language implements multimodality is worth investigating, because it tells us something about the core properties of language.

In this article, we will explore these questions more thoroughly. Specifically, we aim to answer the following research questions: given their consistency across speakers and languages, are there invariant triggers of co-speech gestures? In particular, considering the model outlined in section 1, what information is available in the syntax that the sensorimotor system can interpret and translate into gestures?

¹⁸ For the discussion of the alignment in surprise and surprise-disapproval sentences see Giorgi & Dal Farra (2019), for the cross-linguistic uniformity of gestures see Petrocchi (2022) and Giorgi & Petrocchi (2024). For a general discussion of the theoretical consequences of these findings, see Giorgi & Petrocchi (2025).

Based on the structure provided above, for disapproval special questions introduced by the adversative particle *ma* (but), we can identify three possible triggers on the left of the clausal structure: the speaker's expectations, the interrogative operator, and the empty evaluative projection.

Our research design, therefore, manipulates these three factors: adversativity, interrogativity, and emotionality, comparing them with neutral declarative sentences. We designed a 2^2 experimental plan, contrasting adversative vs non-adversative; interrogative vs non-interrogative and emotional vs non-emotional stimuli. Our experimental questions are the following: do the projections indicated above function as triggers for co-speech gestures? If so, at what level of significance? Are there typical gestures for each different trigger, and which ones? In the next section, we will illustrate the materials and the methodology used in the experiment.

5 Methods and materials

We conducted an experiment with a total of 19 participants (11 females, 8 males; mean age 28.9). Since this study focuses on the gestural expression of emotions, the observed gender imbalance is not expected to bias the results: previous research on facial and bodily expressions of emotion has found no consistent gender differences in emotion display and perception (Cavichio & Sandler, 2015; Fischer et al., 2018). Additionally, our earlier studies on non-lexical co-speech gestures produced in sync with functional positions in syntax showed no influence of gender (Giorgi & Petrocchi 2024; Giorgi & Petrocchi 2025). In our view, this supports the suitability of the present sample concerning the gestural component and the main factors examined, namely emotional expression and the syntactic features of adversativity.

All participants completed a questionnaire about their linguistic background and other potentially confounding variables (e.g., educational level) to allow for the inclusion of these factors as covariates if needed, or to check the homogeneity of the sample. All participants took part on a voluntary basis and provided written informed consent and signed a release form, in accordance with institutional guidelines.

The sample size was determined through an a priori power analysis conducted by simulation in R using the *simr* package (Green & MacLeod, 2016). We modelled the experimental design (19 participants, 56 sentences each, distributed across 8 conditions) within a generalized linear mixed model with a binomial outcome (gesture present/absent). Assuming a medium effect size (odds ratio ≈ 2), $\alpha = 0.05$, and power = 0.80, 1000 simulations were run, indicating that the available sample provides approximately 90% power (95% CI: 88–92%). Thus, the chosen sample size is sufficient to reliably detect effects of the expected magnitude.

5.1 Materials

As outlined above, the stimulus set was designed to investigate the role of gestural triggers depending on three properties: adversativity, interrogativity, and emotionality. Each of these properties corresponds to a syntactic projection: the adversative expectations, the interrogative projection, and the evaluative one. Our design aims at testing the contribution of each of them in triggering gettuality. We have a total of eight conditions. The first four are adversative sentences, in that they are all introduced by the particle *ma*: (A) emotional adversative interrogatives, (B) non-emotional adversative interrogatives, (C) non-emotional adversative declarative sentences, (D) emotional adversative declarative sentences. The emotional sentences all convey a disapproval feeling by the speaker. We selected disapproval, because in our earlier experiments – Giorgi & Dal Farra (2019) and Giorgi & Petrocchi (2024) – gave the clearest effects.¹⁹

The second set mirrored the design of the first, containing identical conditions but utilizing only non-adversative sentences. It included therefore: (A') emotional interrogatives; (B') non-emotional interrogatives; (C') neutral declaratives, i.e., neither interrogative nor emotional; (D') emotional declaratives.

This design allowed us to cross all experimental conditions with appropriate controls, ensuring that the influence of emotional, adversative, and interrogative features could be systematically assessed. The following picture summarizes our experimental design:

Condition	Adversativity	Emotionality	Interrogativity	Sentence Type
A	+	+	+	Emotional, adversative interrogatives
B	+	–	+	Non-emotional adversative interrogatives
C	+	–	–	Non-emotional adversatives declaratives
D	+	+	–	Emotional adversatives declaratives

¹⁹ As note both in Giorgi & Dal Farra (2019) and Giorgi & Petrocchi (2024), surprise is often mixed with a disapproval feeling, presumably due to disappointment for the unexpected situation. Disapproval sentences, are therefore more uniform with respect to the feeling expressed by the speaker.

A'	–	+	+	Emotional interrogatives
B'	–	–	+	Non-emotional interrogatives
C'	–	–	–	Simple declaratives
D'	–	+	–	Emotional declaratives

Table 1: Experimental Conditions

We used a Sentence Repetition Task (SRT) because it has proven to be a reliable and easily adaptable method for experimental purposes. In particular, this methodology is optimal for naturally eliciting target sentences: it enables structured elicitation of specific syntactic structures while maintaining a degree of naturalness, despite the inherently artificial conditions of the experimental setting. This aspect is crucial in our domain of research, where spontaneity plays a central role. In fact, beyond the unavoidable social constraints that may inhibit gesturing (e.g., gesturing is often seen as impolite), previous studies have also shown that artificiality, role-playing, or lack of genuineness in the expression of emotions can significantly impact the study of emotional phenomena (Jürgens et al. 2015; Hoque & Picard 2011).²⁰

We created seven specific contexts for each of the eight experimental conditions. Each context depicts a conversational scenario between two individuals, with one being the participant. In condition A, the scenarios are designed to elicit disapproval, as shown in the example below:

- (52) Context, read aloud in Italian by the experimenter: “You see your younger brother kneeling in the mud in the garden while wearing his best trousers. You think the trousers will be ruined. You are annoyed and say”:
Sentence to be repeated by the subject: “*Ma cosa fai?*” (but what are you doing?)

The utterance in (52) has been defined as a special question – see Giorgi & Petrocchi (2024) for a detailed analysis – namely, as said above, a type of rhetorical wh-question used by the speaker not to

²⁰ To our knowledge, no scientific studies have directly addressed the issue of social constraints that may inhibit gesturing. Nevertheless, indirect evidence can be drawn from the historical debate surrounding the use of sign language in the linguistic rehabilitation of deaf individuals (see, among many others, Fontana 2017). In this context, the use of the term *gesture* to improperly denote a sign is often regarded as pejorative: spoken language is implicitly opposed to what is framed as excessive or inappropriate gesticulation by signers.

request information, but rather to convey a negative emotion, i.e., disapproval.

In condition B, the scenarios are designed to elicit adversative neutral wh-questions, as shown in the example below:

- (53) Context, read aloud in Italian by the experimenter: “You are looking for John and Mary, who arrived together. You see that John is in the living room. You expect Mary to be with him, but you don't see her. You ask:

Sentence to be repeated by the subject: “*Ma Maria dov'è?*” (but where is Mary?).

In this specific case, the questions function as genuine inquiries aimed at eliciting information; however, they are introduced by the adversative particle. This particle explicitly links the utterance in (53) to the speaker's prior, unfulfilled expectations—in this case, the expectation of finding John and Mary together.

In condition C, the scenarios are designed to elicit neutral adversative declarative sentences, as shown in the example below:

- (54) Context, read aloud in Italian by the experimenter: “I tell you that Gianni has decided not to join us on the beach vacation. You know that he has just learned to swim. You say to me”:

Sentence to be repeated by the subject: “*Ma nuotare gli piacerebbe.*” (But he would like swimming.).

Finally, in condition D, the scenarios are designed to elicit emotional adversative declarative sentences, as illustrated in the example below:

- (55) Context, read aloud in Italian by the experimenter: “You and John wanted to go to the cinema. However, John arrives late for the meeting, and the film has already started. You are annoyed and say to him”:

Sentence to be repeated by the subject: “*Ma sei sempre in ritardo!*” (But you're always late!).

The stimuli for the non-adversative set were created using the same criteria, with the only difference being that all sentences are non-adversative. The examples below show scenarios for Condition A' (56), Condition B' (57), Condition C' (58), and Condition D' (59), respectively.

(56) Context, read aloud in Italian by the experimenter: “Your sister was supposed to prepare dinner. However, when you enter the kitchen, you see that she has spilled flour everywhere, the floor is stained with tomato sauce, and there are dirty dishes everywhere. Annoyed, you say to her”:

Sentence to be repeated by the subject: “*Cosa hai combinato?*” (What have you done?).

(57) Context, read aloud in Italian by the experimenter: “You know that Mary went shopping at the new mall. You are curious and ask”:

Sentence to be repeated by the subject: “*Cosa ha comprato Maria?*” (What did Mary buy?).

(58) Context, read aloud in Italian by the experimenter: “Mary wants to know John's hobby. You tell her”:

Sentence to be repeated by the subject: “*Gianni gioca a scacchi.*” (John plays chess.).

(59) Context, read aloud in Italian by the experimenter: “John expresses an opinion on a matter important to you with which you disagree. You are annoyed and say to him”:

Sentence to be repeated by the subject: “*Hai detto una sciocchezza!*” (That's nonsense!).

Two stimulus lists were created, both with stimuli randomized across conditions and sentence types. No two stimuli of the same type were ever presented consecutively.²¹

5.2 Procedure

Participants were instructed to stand in a relaxed posture, with their arms naturally resting at their sides, approximately 40 cm from the computer screen. The experiment began with a background questionnaire designed to gather information about the participant's linguistic history. This phase was conducted with the camera already turned on to promote comfort and a natural demeanor in front of the camera. The actual task only started after a brief, informal conversation between the experimenter and the participant, allowing the latter time to settle into the experimental setting.

After listening to the context, read aloud by the experimenter, the participant was shown a target sentence to repeat. Stimulus presentation was controlled by a PC running PowerPoint. The sentence

²¹ To prevent fatigue effects associated with the total number of items (56 stimuli), the participants were divided into two groups: one group received List 1 and the other List 2. In List 2, the order of the items was reversed with respect to List 1, such that the last 28 stimuli of List 1 became the first 28 stimuli of List 2. This counterbalancing procedure ensured that no subset of stimuli was systematically presented at the end of the experiment, thus avoiding potential position-based penalization.

appeared on the screen for a few seconds, during which the participant read it silently. Once the stimulus disappeared and a blank slide appeared, serving as a visual cue, the participant was expected to repeat the sentence aloud from memory, imagining themselves in the previously described scenario. The instruction was to repeat the sentence as naturally as possible, delivering the utterance as if it were part of a real conversation. We also never corrected the participant if the repeated sentence was not exactly the same as the target stimulus.²²

The instructions given to the participants contained no references to gestural cues, nor were any non-verbal directions provided during stimulus presentation or in the written contexts. This approach ensured that the production of both manual and non-manual gestures was spontaneous and solely driven. The sentences to be repeated, shown in written form, included punctuation marks (e.g., question marks, exclamation points). This decision was made to preserve maximum naturalness in the speakers' production. In fact, a pilot phase showed that displaying neutral questions without punctuation made participants feel uncomfortable, leading to a loss of naturalness and difficulty in repeating the sentences without visual cues. Therefore, to maintain consistency, all stimuli were adapted and presented with punctuation. The reason for this choice is that our study aimed to compare the gestural realization of the sentences and was not focused on assessing the interpretation participants attributed to them, i.e., their emotional value; to this end, the task explicitly prompted the target interpretation aligned with the experimental design.

5.3 Data Transcription and Analysis

The participants were videotaped, and the footage was analyzed. The video was segmented using Kdenlive 23.04.1, and only the segments where the speaker repeated the target sentence were selected. Each participant contributed 56 clips, one for each of the 56 target stimuli, totaling 1,064 stimuli ($56 \text{ stimuli} \times 19 \text{ participants}$). The video data were transcribed and annotated with ELAN version 6.9. To ensure precise timing and synchronization with the linguistic material, co-speech gestures were annotated and aligned with the video, using a template with tiers for manual gestures, head movements, eyebrow movements, shoulder movements, eye aperture, torso movements, speech transcription, and additional notes. The main goal of the ELAN analysis was to identify both manual and non-manual gestures and explore their potential alignment with the linguistic material. The

²² Actually, these repetition “errors” constitute an interesting source of information, as noted in our previous work, particularly in the study of warnings (Giorgi & Petrocchi, 2025a), where several dialectal items entered the repetition task, due to the peculiar nature of the stimuli, which included interjections. Interjections were often substituted by dialectal ones. We will further analyze these special productions in future work.

annotated data were then extracted in long format and compiled into a summary Excel spreadsheet for statistical analysis. We coded the gestures produced by participants while they repeated the target stimuli. Our analysis distinguished between manual and non-manual gestures, focusing primarily on the stroke phase for both.

5.3.1 Manual Gestures

Regarding manual gestures, our analysis centered on the stroke phase, which we consider the most important part of the movement. The Stroke Phase is the crucial moment when the main movement occurs and the gesture assumes its complete meaning and definitive form (Kendon 2004). Subsequent studies have also demonstrated the central importance of the gesture stroke in analyzing its co-alignment with the co-occurring linguistic element and the relevant correlated prosodic pitch (Seyfeddinipur, 2006; Sowa, 2006).²³

The entire gesture unit (preparation phase, stroke, and hold phase) was considered to represent the full scope of the gesture within the sentence (Kendon 2004; Bressemer & Ladewig, 2011).

5.3.2 Non-Manual Gestures

For non-manual gestures, we coded the stroke as the first complete occurrence of the gesture itself (Church et al., 2017). This includes, for example, the full eyebrow movement, eye opening, head tilt, and forward torso movement when they first reach their definitive position. In the next section, we will explain in more detail the gestures taken into account.

5.4 Gestural Types and Inclusion Criteria

Only gestures attested in at least 7 out of 19 participants ($\approx 40\%$) were considered robust enough to be included in the analysis. This threshold was chosen to balance sensitivity in detecting shared multimodal patterns with control over individual idiosyncrasies.

Among the non-manual gestures, *furrowed eyebrows* (17 participants, 172 occurrences) and *side-to-side head movement* (18 participants, 180 occurrences) exceeded this threshold. Among the manual gestures, three types were included: the *palm-up open hand (PUOH)* gesture (19 participants, 138 occurrences), the *artichoke* gesture (7 participants, 29 occurrences), and the *hands-in-prayer* gesture (8 participants, 63 occurrences).

²³ Data concerning the prosodic production of the sentences were also collected. The second phase of this research will involve the prosodic analysis of the utterances.

The side-to-side head movement corresponds to what Kendon (2002; 2004) describes as a gesture distinct from the head shake of negation, which typically consists of one or two rapid, small-amplitude horizontal rotations functioning as an explicit negator and a conventionalized emblem in Western cultures (McClave 2000; Kendon 2002). By contrast, the side-to-side movement involves wider and slower repetitions, often performed more than twice, and expresses an evaluative or emotional stance, such as disapproval (Kendon 2002:152). In Kendon's corpus, for instance, such a movement appears in a speaker discussing why Neapolitans feel ashamed of being from Naples. The speaker shakes her head slowly uttering the sentence "*C'è troppa delinquenza*" ("There is too much crime"). This gesture can be interpreted as an expression of a negative attitude towards what is being said: there is no direct verbal equivalent, but the gesture clearly conveys a negative evaluation toward the propositional content (Kendon 2002:167).²⁴

Furrowed eyebrows involve lowering and inward movement of the brows, creating vertical wrinkles between them. In the literature, they are linked to cognitive effort, evaluative stance, and negative emotions such as anger or disapproval (Ekman & Friesen, 1969; Ekman, 1993; Kendon, 2004; Bavelas & Chovil, 2018). Recent research has shown that eyebrow movements play a functional role in signaling communication issues in spoken face-to-face interactions (Hömke et al., 2025). In particular, speakers are sensitive to the addressee's eyebrow furrowing as a signal of insufficient understanding, as demonstrated by their tendency to give longer answers when interacting with a brow-furrowing listener. This supports the hypothesis that addressee eyebrow furrowing is interpreted as a cue of comprehension difficulty. Moreover, the authors propose that furrowing the brows may also reflect an internal processing difficulty or a high cognitive load on the part of the addressee, which the speaker then interprets as a need for clarification.

Interestingly, eyebrow movements have also been proposed to occur in requests for information directed at a conversational partner (Ekman & Friesen, 1975; Wierzbicka, 1991) and are clearly grammaticalized as facial question markers in many sign languages (Zeshan, 2004; Dachkovsky & Sandler, 2009; Pfau et al., 2012). Specifically, in some sign languages, such as Italian Sign Language (LIS) and Israeli Sign Language (ISL), furrowed eyebrows serve to mark *wh*-questions.

Among manual gestures, the *Palm-Up Open Hand* (PUOH) gesture consists of one or both hands held palm-up and slightly extended forward with a relaxed open posture. It has been widely discussed in the literature as a conventionalized form associated with offering, showing, or presenting something to the interlocutor (Kendon 2004; Müller 2004). This gesture family is considered a recurrent gesture type that expresses openness and appeal within interaction (Cienki 2017). Recent

²⁴ Note that the presence of this gesture is coherent with the emotional value of the experimental stimuli, i.e., disapproval.

accounts further emphasize its role as an epistemic and interactive gesture, marking the speaker's stance and engagement in dialogue (Müller, Bressem & Ladewig 2013; Bressem & Müller 2017). The *artichoke* gesture consists of one or both hands held in front of the torso, with the fingers loosely grouped together in a concave, “pinched” configuration — resembling the shape of an artichoke. This gesture has received considerable attention in studies on Italian gesturality, where it is described as a culturally emblematic form (Poggi 1983; Kendon 1995, 2004). It conveys meanings related to emphasis, contrast, or evaluative stance, depending on prosodic and contextual alignment. Traditionally, it has been depicted as a gesture used for asking a question (Munari 1963; Diadori 1990). According to Kendon (2004: 231–232), in Neapolitan varieties, this gesture typically occurs when the speaker is confronted with something that challenges expectations or understanding, demanding an explanation or justification. It is used when the speaker asks a question because they are surprised, annoyed, or puzzled by something, or when they are testing the interlocutor's knowledge, but not when merely requesting information. Kendon also notes that the gesture can be accompanied by an oscillatory movement, usually in the vertical plane, which highlights the object of discourse. Poggi (1983) describes this gesture as typical of “negative pseudo-questions,” in which the speaker does not genuinely seek information but rather questions and implicitly criticizes the interlocutor — as in “What are you saying?” or “What are you doing?” in a tone of disapproval or surprise. Finally, a corresponding sign has been identified in Italian Sign Language (LIS), known as *QARTICHOKE* (Branchini et al. 2013). This sign, formally resembling the *artichoke* gesture, functions as a generic wh-element that appears only in wh-questions, often in clause-final position, serving as an underspecified interrogative marker.

The *hands-in-prayer* gesture, involving both palms pressed together and held in front of the chest or face, has been described as an emblematic form (Poggi, 1983; Kendon, 1995) and has been particularly observed in the Salerno area (Kendon, 1995). Traditionally, this gesture has been interpreted as expressing a plea for indulgence, reflecting the idea that the hands are metaphorically bound and therefore unable to act (De Jorio, 1979). According to Kendon (1995), it can also denote the impossibility of performing an action, as in the expression “*What can I do?*” (*Cosa posso farci?*), showing helplessness or resignation.

More recently, these gestures have been observed in studies on surprise and disapproval (Giorgi & Dal Farra, 2019; Giorgi & Petrocchi, 2024). PUOH, artichoke gesture, and hands-in-prayer gesture appeared consistently in *wh*-questions expressing surprise and disapproval in Italian, in experimental settings and designs similar to the one used in the present study. Comparable patterns have been reported for other languages such as Spanish (Furlan, 2019), Japanese, and Vietnamese (Petrocchi, 2022; Giorgi & Petrocchi, 2024).

These manual gestures were almost always produced in an iterated form — that is, repeated two or more times through slight upward and downward movements. Iteration does not define a distinct gesture type but rather a mode of realization that modulates expressive or pragmatic force. In the literature, repetition has been linked to aspectual meaning (iteration or continuity), intensification, and emotional engagement (McNeill, 1992; Ladewig, 2014). It also recalls Kendon’s description of a vertical oscillatory movement frequently accompanying the artichoke gesture to enhance salience. Indeed, iteration distinguishes the disapproval-PUOH observed in our study from other gestures in the PUAH family, which have been characterized as representational or interactive gestures of offering, giving, or requesting engagement, typically conveying epistemic or attitudinal meanings such as openness, uncertainty, or appeal. Similarly, the artichoke and hands-in-prayer gestures are obligatorily associated with iteration.

It is worth noting that, while the PUAH gesture was used by participants for sentences expressing disapproval — and has been reported cross-linguistically (e.g., in Italian, Spanish, Japanese, and Vietnamese; Giorgi & Dal Farra, 2019; Furlan, 2019; Giorgi & Petrocchi, 2024; Petrocchi, 2022) — the artichoke and hands-in-prayer gestures were produced by fewer speakers and only in Italian and Spanish.²⁵ These two gestures display similar distributions across experimental conditions, both occurring, for example, in adversative emotional sentences.²⁶

Finally, we have to note that previous literature on artichoke gesture and hands in prayer gesture has described them as illocutionary gestures—that is, gestures related to sentence type, clause force, or the evaluative domain (Poggi, 1993; Kendon, 1995). These authors intuited the syntactic nature of such gestures, recognizing — *de facto* - their role at the left periphery of the clause structure.

It should also be noted that some gestures, although exceeding the inclusion threshold, were excluded from the analysis for independent methodological reasons. For example, raised eyebrows occurred in 14 of the 19 participants (96 occurrences), but more than half of these instances (51%) appeared in non-interrogative, non-emotional sentences (both adversative and non-adversative). A closer examination of gesture–speech alignment revealed that raised eyebrows were often synchronized with given information within the utterance, marking topical specification rather than expressive stance. For instance, consider the following example:

²⁵ However, recent studies on multimodal constructions in German have shown that the artichoke gesture also occurs in this language. In particular, Shoonjans (2018) observed that this gesture frequently accompanies evaluative constructions involving the modal particle *doch*, suggesting that its use extends beyond Italian cultural conventions and may be tied to broader pragmatic functions related to contrastivity and evaluation.

²⁶ Preliminary observations on our Italian data suggest a possible geographic variation in the use of these gestures, which will require further analyses to confirm. A detailed investigation of this issue is left to future work.

- (60) Context, read aloud in Italian by the experimenter: “A friend asks you how long the movie you watched last night lasted. You answer.”:
Sentence to be repeated by the subject: “*Il film è durato due ore* ” (“The movie lasted two hours”)

In this context (Condition C’, non-adversative, non-emotional, non-interrogative sentence), speakers tended to raise their eyebrows on *il film* (“the movie”), highlighting it as given or topical information rather than as an expression of surprise or evaluation. While this issue falls beyond the scope of the present paper, it clearly warrants further investigation in future work.

Two additional gestures – the head shake (10 participants, 14 occurrences) and the palm-down gesture of negation (12 participants, 39 occurrences) – also exceeded the frequency threshold but were excluded because they are semantic or emblematic gestures rather than non-semantic co-speech gestures, which are the focus of this study. The head shake of negation is considered an emblem in Western cultures (McClave, 2000; Kendon, 2002), as it can directly replace the verbal particle *no* and convey the same propositional meaning. Similarly, the palm-down gesture of negation was excluded from the present analysis. Following Harrison (2024), this gesture can be regarded as a conventionalized pragmatic form at the boundary between emblematic and interactional gestures. Since the present study focuses on non-semantic co-speech gestures, the palm-down gesture was therefore excluded from the analysis.

6 Results

The following analyses were designed to quantitatively evaluate the relationship between macro-linguistic factors and gestural behavior across various dimensions of multimodal expression. We counted the gestures produced by participants in each condition, categorizing them according to the typology outlined in section 5.4.

Given the nature of the data, two complementary statistical approaches were adopted. First, a Negative Binomial Generalized Linear Mixed Model (GLMM) was fitted to the *Gesture Count* variable, representing the total number of gestures produced per stimulus. This model is appropriate for over-dispersed count data and allowed us to test whether the three macro-linguistic factors and their interactions significantly predicted the overall frequency of gestural production.

Second, a series of binomial (logistic) GLMMs was performed on *binary outcome variables* indicating the presence or absence of specific gesture types (Furrowed Eyebrows, Side-to-side Head

gesture, and Hand gestures) or gestural features (Iteration). This modeling strategy was chosen to capture the probabilistic nature of type-specific gesture occurrence, independently of their absolute frequency.

6.1 Negative Binomial GLMM: Gesture Count

To test the hypothesis that the presence of the three macro-linguistic factors increases gestural output, a Negative Binomial Generalized Linear Mixed Model (GLMM), suitable for over-dispersed count data, was performed on the *Gesture count* variable. The model included the three macro-groups and their interactions as fixed effects, with Participant and Stimulus as random intercepts to account for repeated measures and baseline variability. The analysis confirmed significant main effects for all three factors (Type III Wald Chi-square tests): Emotionality ($\chi^2 = 32.37$, $p < 0.001$), Interrogativity ($\chi^2 = 11.21$, $p < 0.001$), and Adversativity ($\chi^2 = 24.74$, $p < 0.001$).

These results show that each macro-group independently associates with a significant increase in the expected number of gestures. A significant two-way interaction was found between Emotionality and Adversativity ($\chi^2 = 14.52$, $p < 0.001$). Post-hoc analysis using Estimated Marginal Means (EMMs) revealed that the facilitatory effect of Emotionality on gesture count was highly significant when Adversativity was absent (Rate Ratio, $RR = 3.07$, $p < 0.0001$), but this effect was attenuated when Adversativity was present ($RR = 1.20$, $p = 0.096$). Importantly, this attenuation in additivity does not indicate a reduction in overall gestural activity under co-occurrence. Consider Figure 2:

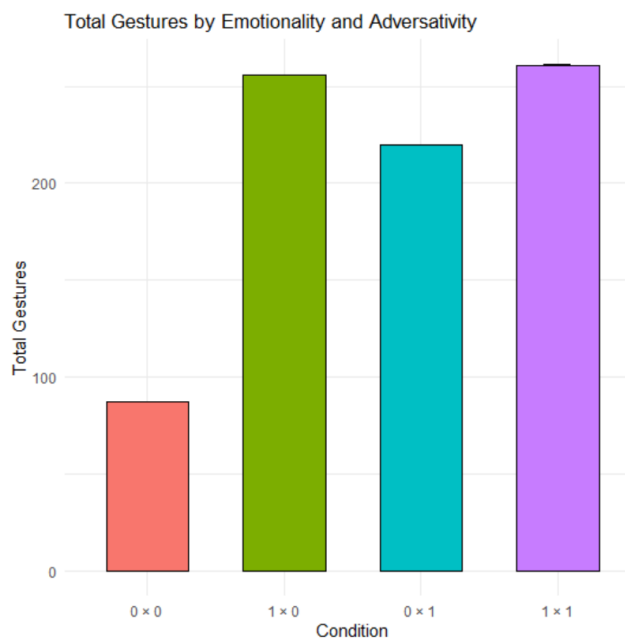


Figure 2: Gestural output by Emotionality × Adversativity condition.

The graph in Figure 2 illustrates the total number of gestures observed across different *Emotionality* and *Adversativity* conditions. Both the 1×0 condition — the green one, representing *Emotionality* — and the 0×1 condition — the blue one, representing *Adversativity* — indicate that each factor alone increases the total number of gestures compared to the 0×0 baseline — the red one, representing the absence of both factors. The interaction between *Emotionality* and *Adversativity* does not seem purely additive, as the combined 1×1 condition — the violet one — does not equal the sum of the single-factor totals (1×0 plus 0×1). Nonetheless, this condition still results in a slightly higher total number of gestures than 0×1 , and 1×0 , suggesting that the co-occurrence of both factors boosts gestural activity. Additionally, the 1×0 condition produces more gestures than 0×1 , suggesting that *Emotionality* alone triggers more gestures than *Adversativity* in isolation.

Overall, the pattern reflects a complex, non-linear interaction: *Emotionality* alone elicits more gestures than *Adversativity* alone, and the non-additive pattern they create does not indicate a decrease in overall gestural activity when both factors are present, as can be seen by the slight increase in gestures. We will further discuss this outcome in section 7 below.

6.2 Logistic GLMM: Gesture Type-Specific Analysis

To examine which macro-linguistic factors influence the likelihood of producing specific gesture types (Furrowed eyebrow gesture, Head gesture, Hands gesture, Iteration), four separate Generalized Linear Mixed Models with a binomial (logistic) distribution were conducted. The results revealed a functional dissociation among the gesture categories.

6.2.1 Furrowed Eyebrows Gesture

This gesture was significantly increased by both Interrogativity ($\chi^2 = 18.38$, $p < 0.001$) and Adversativity ($\chi^2 = 12.22$, $p < 0.001$), with both factors also entering into significant two-way interactions ($\chi^2 = 4.24$, $p < 0.040$).

The main cause of Furrowed Eyebrows production was confirmed to be Interrogativity, which raised the baseline probability of the gesture from 2.60% to 20.93% (when *Emotionality* was absent). The significant effect of Interrogativity corresponds to an Odds Ratio (OR) of approximately 9.91 ($p < 0.001$).

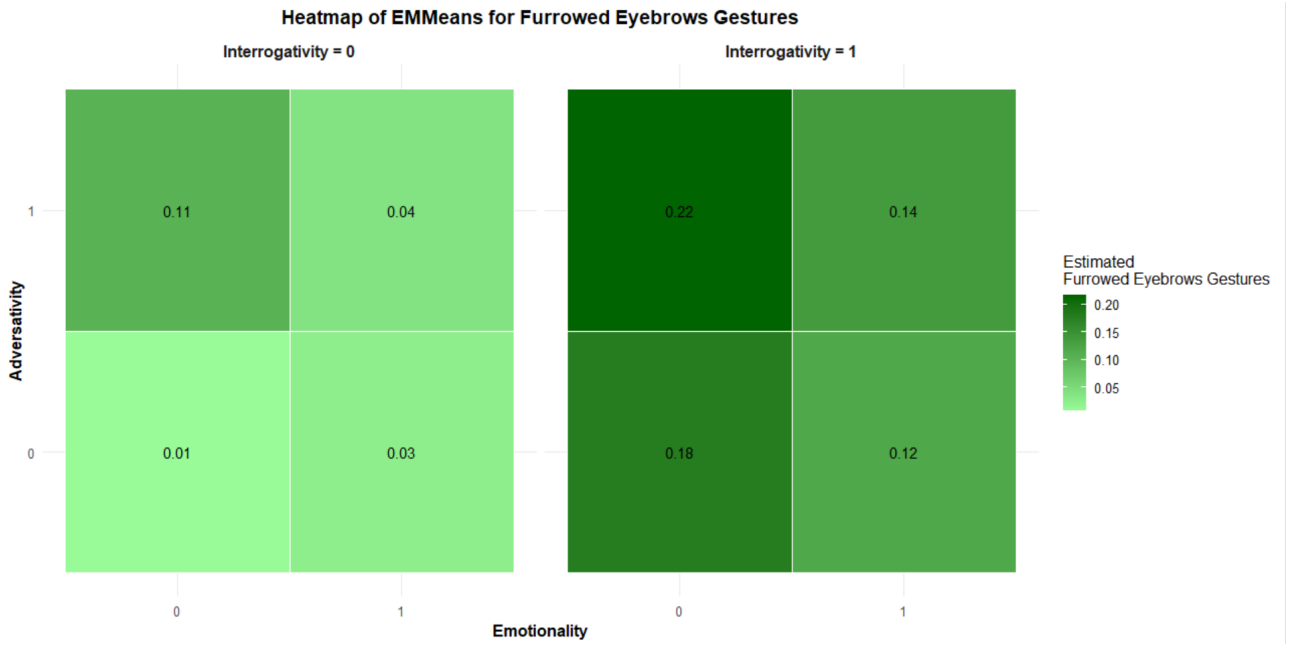


Figure 3: Estimated marginal means for Furrowed Eyebrows gestures across Emotionality and Adversativity, faceted by Interrogativity.

The heatmap in Figure 3 visualizes the estimated marginal means (EMMeans) of Furrowed Eyebrows gestures across all combinations of Emotionality, Adversativity, and Interrogativity. Each tile represents the predicted number of gestures for a given combination of Emotionality (x-axis) and Adversativity (y-axis), with separate facets for Interrogativity (0 = absent, 1 = present). The color intensity corresponds to the estimated gesture rate, with darker green indicating higher predicted values.

As expected, Interrogativity appears to be the primary driver of Furrowed Eyebrows production: the facet where Interrogativity = 1 shows substantially higher rates than the Interrogativity = 0 facet, even when Emotionality and Adversativity are absent. Adversativity has a secondary, facilitative effect, particularly in the presence of Interrogativity, while Emotionality has minimal influence. The numeric labels on each tile indicate the exact estimated gesture counts, making it easy to see the magnitude of predicted differences. Overall, the heatmap confirms that Furrowed Eyebrows gestures are mainly triggered by interrogative contexts, with Adversativity enhancing this effect.

6.2.2 Head Gesture

Head gestures were strongly and uniquely driven by Emotionality ($\chi^2 = 22.19$, $p < 0.001$), entering into a significant three-way interaction.

To precisely interpret this interaction ($\chi^2 = 4.08$, $p = 0.043$), a comprehensive post-hoc analysis using EMMs was conducted, focusing on the conditional effect of Emotionality.

Emotionality is the primary trigger of Head gestures ($OR \approx 14.18$, $p < 0.0001$ in the baseline context Interrogativity = 0, Adversativity = 0). This effect reached its maximum potentiation when Adversativity was present and Interrogativity was absent ($OR \approx 24.81$, $p = 0.0001$), confirming Adversativity as a synergistic enhancer.

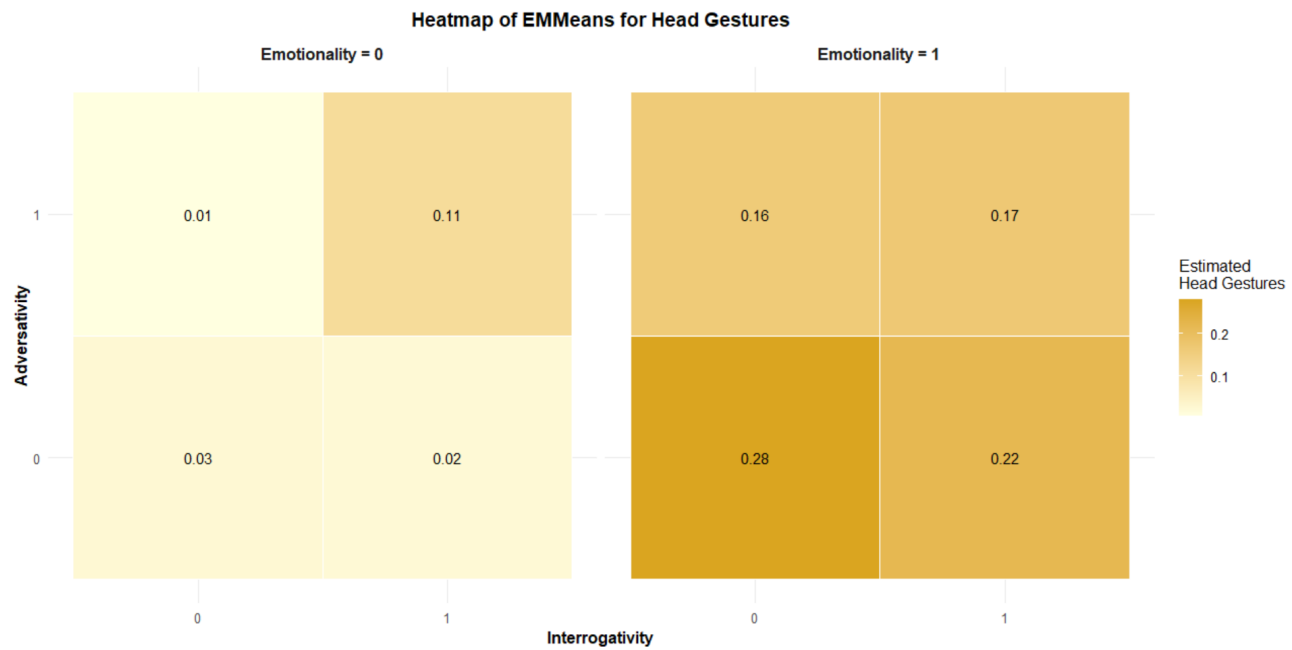


Figure 4: Estimated marginal means for Head gestures across Adversativity $\times$ Interrogativity conditions, faceted by Emotionality.

The heatmap in Figure 4 displays the estimated marginal means (EMMeans) of Head gestures across combinations of Adversativity and Interrogativity, with separate panels for Emotionality = 0 and Emotionality = 1. Each tile represents the predicted probability of producing a Head gesture for a given combination of Adversativity (y-axis) and Interrogativity (x-axis), with color intensity corresponding to the estimated probability (darker yellow = higher probability). Numeric labels on the tiles indicate the exact predicted probabilities.

Emotionality is the primary driver of Head gestures: the panel for Emotionality = 1 shows substantially higher probabilities across nearly all combinations than the panel for Emotionality = 0. Interrogativity and Adversativity act as modulators: the effect of Emotionality is strongest when Adversativity is present and Interrogativity is absent, reflecting the synergistic enhancement observed in the post-hoc analysis ($OR \approx 24.81$, $p = 0.0001$). Overall, the heatmap clearly illustrates that Head gestures are mainly triggered by emotional contexts, with secondary effects of adversativity shaping the exact probability of production.

6.2.3 Hands Gesture

Hand gestures were significantly facilitated by both Emotionality ($\chi^2 = 9.30$, $p = 0.002$) and Adversativity ($\chi^2 = 15.35$, $p < 0.001$), with a significant interaction between these two factors ($\chi^2 = 4.24$, $p = 0.040$).

To interpret this interaction, a post-hoc analysis using EMMs was performed. The primary promoter of Hand gestures was identified as Emotionality. When Adversativity (Adv = 0) was absent, Emotionality significantly increased the odds of producing Hand gestures by a factor of 6.94 ($p < 0.0001$). When Adversativity was present (Adv = 1), the effect of Emotionality was reduced to a factor of 1.57 ($p = 0.047$). Adversativity alone raised the baseline probability of Hand gestures from 3.05% (Emo = 0, Adv = 0) to 15.58% (Emo = 0, Adv = 1), indicating a non-additive saturation effect.

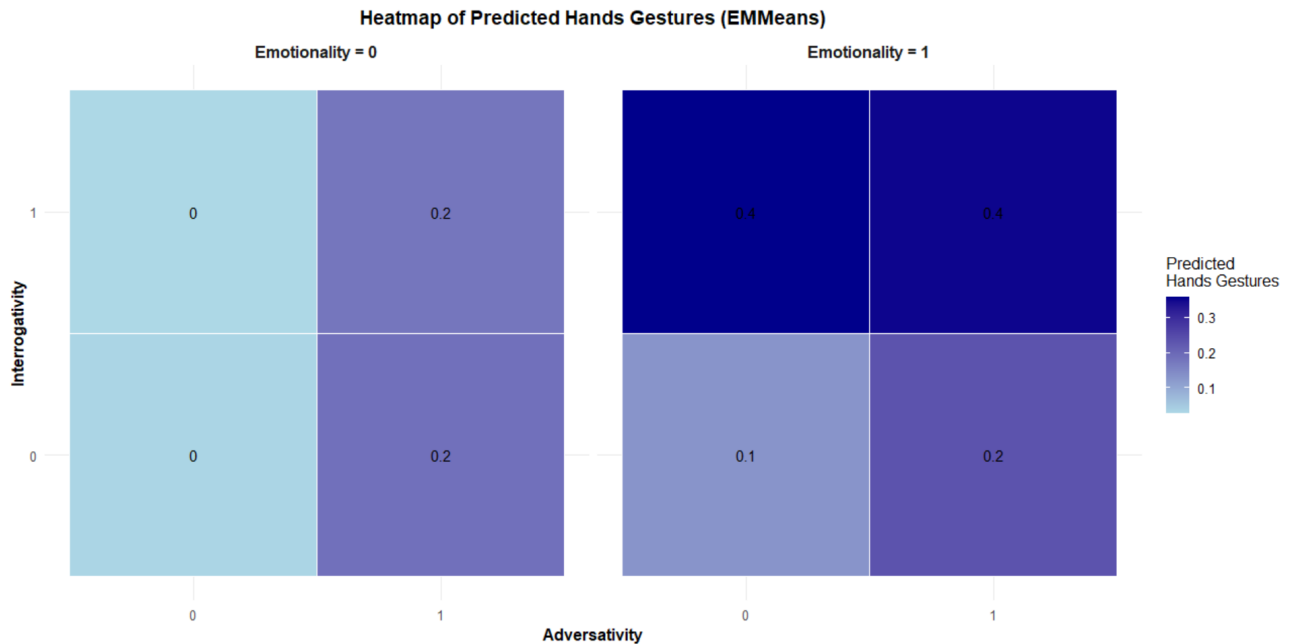


Figure 5: Estimated marginal means for Hands gestures across Adversativity and Interrogativity, faceted by Emotionality.

The heatmap in Figure 5 shows the predicted counts of Hands gestures (derived from EMMeans) across combinations of Adversativity (x-axis) and Interrogativity (y-axis), with separate panels for Emotionality = 0 and Emotionality = 1. Each tile represents the estimated number of gestures for a specific combination, with color intensity reflecting the predicted counts (darker blue = higher counts) and numeric labels showing the exact values.

As illustrated, Emotionality is the primary trigger for Hands gestures: the panel for Emotionality = 1 consistently shows higher predicted counts than the panel for Emotionality = 0. Adversativity

modulates the effect of Emotionality, increasing the predicted gestures, especially when Emotionality is absent, indicating a non-additive enhancement. Interrogativity has a smaller, conditional effect, slightly changing the predicted counts depending on the combination. Overall, the heatmap reveals a synergistic interaction: Emotionality strongly promotes Hands gestures, with Adversativity amplifying this effect under specific conditions.

6.2.3 Iteration

The probability of producing Iteration was not significantly affected by any single main effect ($p > 0.05$), but was instead determined by the three-way interaction of Emotionality, Interrogativity, and Adversativity ($\chi^2 = 5.24$, $p = 0.022$).

Post-hoc EMMs analysis revealed that Emotionality is a highly conditional promoter of Iteration. The strongest facilitatory effect occurred when Interrogativity was present and Adversativity was absent ($\text{Int} = 1$, $\text{Adv} = 0$): in this condition, Emotionality significantly increased the odds of producing Iteration gestures by a factor of approximately 43.3 ($p < 0.0001$). Emotionality also served as a significant promoter when Adversativity was present and Interrogativity was absent ($\text{OR} \approx 11.6$, $p = 0.004$).

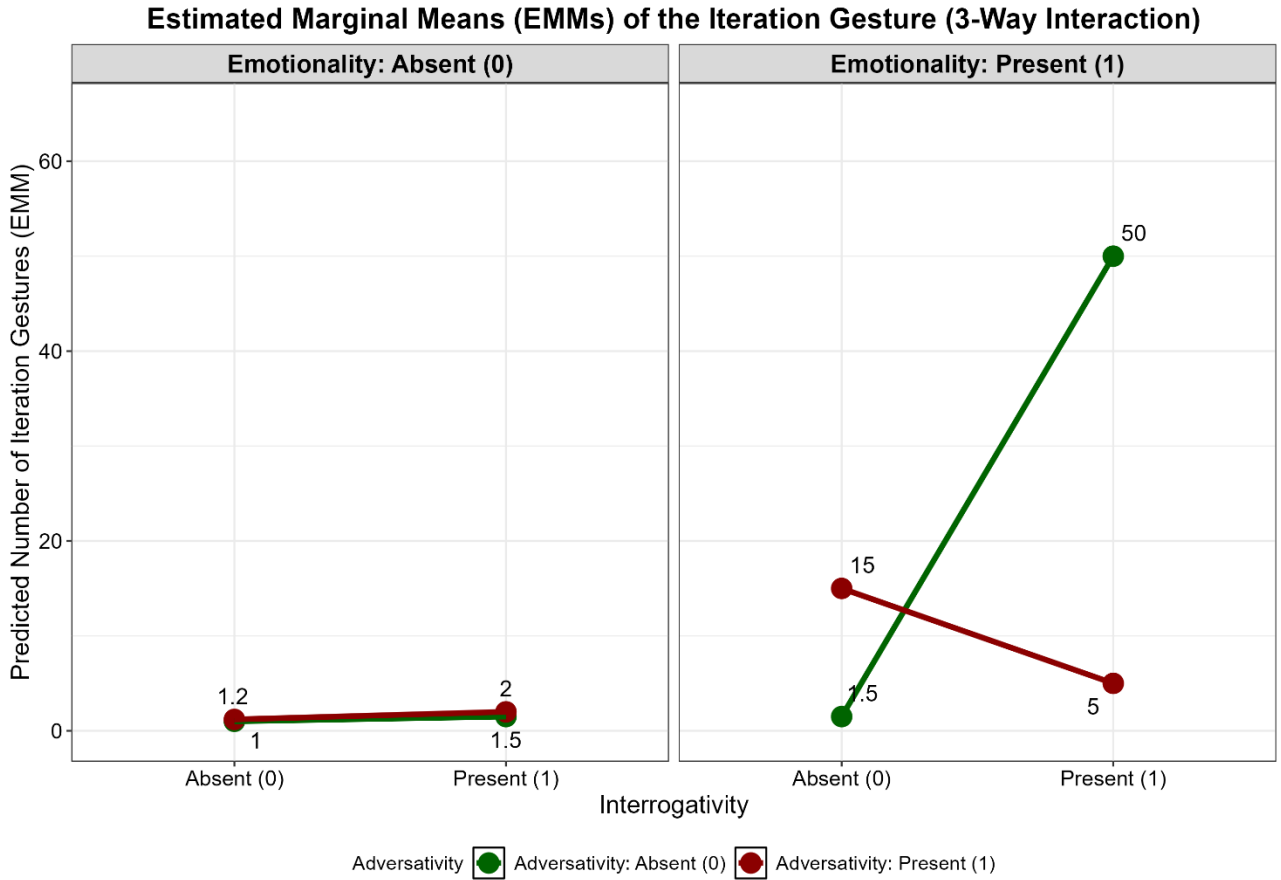


Figure 6: Estimated marginal means for Iteration gestures across Interrogativity and Adversativity, faceted by Emotionality

The interaction plot in Figure 6 shows the estimated marginal means of Iteration gestures for all combinations of Emotionality, Interrogativity, and Adversativity. The y-axis represents the predicted number of Iteration gestures, transformed from the log scale to ensure interpretable, positive values. Each panel represents one level of Emotionality (0 = absent, 1 = present), while lines and colors differentiate Adversativity (0 = absent, 1 = present). The x-axis displays Interrogativity (0 = absent, 1 = present).

The figure shows that Iteration gestures are highly conditional, consistent with the significant three-way interaction observed in the GLMM ($\chi^2 = 5.24$, $p = 0.022$). Emotionality acts as a strong facilitator only under specific contexts: its effect is maximal when Interrogativity is present and Adversativity is absent, and substantial when Adversativity is present and Interrogativity is absent. In contrast, when both Interrogativity and Adversativity are absent, Emotionality exerts a minimal influence. This pattern highlights that Iteration gestures are not driven by any single factor in isolation, but rather emerge from the interplay among the three macro-linguistic variables.

7. Discussion

In this section, we will address the research questions posed in section 4 regarding the relationship between the syntactic representation and the sensorimotor component, based on the statistical results outlined above.

7.1 The syntactic projections as triggers for gestuality

The statistical results discussed in section 6.1 show that each factor – adversativity, emotionality, and interrogativity – produces significant effects. Each one, when considered alone, acts as a trigger to increase gestuality compared to neutral conditions, i.e., participants produce more gestures when one of these triggers is present in the stimulus.

Therefore, the first of our research questions receives a positive answer. Each of the projections we hypothesized in the syntactic representation, namely the unexpressed expectations, the interrogative projection and the evaluative one, triggers co-speech gestures at a high level of significance. Furthermore, based on our statistical results, the most significant factor is emotionality, followed by adversativity, and then interrogativity.

Although all three factors trigger gestures, the link between emotionality and adversativity is more intricate. Notably, our findings indicate that their combined effect is not simply additive, in that the total gestures produced when both factors are present are fewer than the sum of the gestures in each separate condition.

This can be easily understood by looking at the gestures involved in the different conditions. Clearly, if the same body part is used in both cases – as shown by the statistical analysis in section 6.2, which we will review shortly – we cannot expect the total number of gestures to be equal to the sum of those in each individual condition. The only way the number of gestures can increase when they involve the same body part is this: if a participant *did not* gesture when only one factor was present, then when two factors are present, the likelihood that the same participant gestures is higher. This results in a higher total number of gestures in the additive condition compared to the single condition. However, the number of participants who *did not* move in either emotional or adversative sentences is very small. Therefore, there is no *significant* additive effect, precisely because the two conditions tend to trigger gestures involving the same body part. Note, however, that there is indeed a slight increase in the number of gestures when both conditions are present, but it is not statistically significant, as pointed out above.

Summarizing, emotionality and adversativity are the primary factors triggering gestuality, followed by interrogativity, which also exhibits significant activation levels, though lower. The simultaneous

presence of emotionality and adversativity increases gestuality, even if the effect is not conspicuous due to physical limitations.

7.2 The syntactic projections trigger different gestures

In section 6.2, we also examined the single gestures along with the iterativity feature of manual gestures. We found that they do not have the same significance level across different conditions. This is especially interesting because it highlights the importance of our second research question: our data indicate that different syntactic projections trigger different gestures.

The non-manual gesture of *furrowed eyebrows* is mainly triggered by interrogativity, i.e., by the projection of the head INT. As discussed earlier in section 5.4, this gesture is robustly associated with interrogativity in multiple ways. It has been shown to relate to cognitive effort, serving a functional role to signal communication problems during spoken face-to-face interactions. Specifically, it's mostly seen as a sign of understanding difficulty, indicating either internal processing issues or a high cognitive load on the part of the listener, which the speaker interprets as a need for clarification. Eyebrow movements have also been proposed to occur in requests for information directed at a conversational partner (Ekman & Friesen, 1975; Wierzbicka, 1991) and are clearly grammaticalized as facial question markers in many sign languages (Zeshan, 2004; Dachkovsky & Sandler, 2009; Pfau et al., 2012).

Our two-factor analysis revealed that there is a synergistic effect of Adversativity $\times$ Interrogativity that increases the likelihood of producing this gesture, whereas Emotionality does not play a significant role, either alone or in interaction with other factors (see section 6.2.1). This suggests that furrowed eyebrows are not directly triggered by a negative emotion, but rather by a lack of understanding, which can be combined with the contrast introduced by adversativity, i.e., by the fact that the speaker's expectations are not met.

Side-to-side head movement is triggered by emotionality – syntactically realized as the projection of the head EVAL – as shown by the statistically significant levels illustrated in section 6.2. This gesture is linked to the negative evaluation, i.e., disapproval, expressed by these sentences.

Regarding *hand gestures*, it should be noted that, consistently with our previous studies (Giorgi and Dal Farra, 2019; Petrocchi, 2022; Giorgi & Petrocchi, 2024), all participants produced the PUOH gesture at least once; moreover, some participants also produced either the artichoke gesture or the hands-in-prayer gesture in some sentences. We therefore treated the three gestures as belonging to a single category, as different realizations of a gesture type sharing the same function. Indeed, hands-in-prayer and artichoke gestures have previously been classified within the same category, serving

the same evaluative function (Kendon, 1995), i.e., expressing the speaker's opinion in a negative or critical way, sometimes even in the sense of "being at a loss for words."

Both emotionality and adversativity are significant triggers for these gestures. The syntactic projections involved are therefore the one including the *speaker's expectation*, present in adversative special questions, and the EVAL projection.

Finally, *iteration* plays a crucial role: it is primarily triggered by emotionality, i.e. EVAL, but is also strongly influenced by interrogativity. This finding is particularly noteworthy, as iteration characterizes the manual gestures in question, which have consistently been described as typical of rhetorical or pseudo-questions, combining the expression of a negative stance with a questioning form. Most examples in the literature illustrate speakers producing pseudo-questions, i.e., *special questions* in our terminology. In some instances, the violation of expectations is so pronounced that no explanation can be imagined; in such cases, the hands-in-prayer gesture is used precisely to convey this sense of disapproval and "being at a loss for words".

Concluding, each empty projection, present in the syntax to convey the correct interpretation, is also read off by the sensorimotor component and expressed through specialized co-speech gestures in a systematic and statistically significant way.

8 Conclusions

Based on the analysis above, we can state that co-speech gestures are systematically triggered by the syntactic properties of the utterance. Moreover, each projection is specialized in triggering specific gestures. The interrogative projection primarily elicits furrowed eyebrows, a gesture commonly associated with interrogative contexts; the evaluative projection mainly triggers side-to-side head movement, often indicating disapproval; unmet expectations, i.e., adversativity, together with the evaluative projection, trigger hand gestures, with iteration being a characteristic feature. Note that, as discussed earlier, unmet expectations and disapproval are closely connected, as the speaker's disapproval feeling stems precisely from the fact that their expectations are not met.

Therefore, these projections, even if they do not correspond to lexical items, are indeed realized by the sensorimotor component, not only as intonation, as noted in previous works, but also as gestures. Oral language privileges the phonetic output, and gestures are not an obligatory part of sentence realization. Our work, however, clearly shows that gestures, though not obligatory, are not a random and occasional component accompanying utterances, but are a systematic and predictable realization of syntactic projections.

In a word, our observations point to a fundamental multimodal architecture of language, where the alignment of syntax, prosody and gesture constitutes a basic property of the language model.

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